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# Cultivating a Greener Future by A Blue Carbon Breakthrough: The ARLI-CAIA Partnership on Seaweed Carbon Trading

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ABSTRACT:

This study explores the vital role of blue carbon ecosystems, particularly seaweed, in mitigating climate change and facilitating carbon trading. Blue carbon refers to the carbon captured by oceanic and coastal ecosystems, which play a crucial role in sequestering atmospheric CO2. Seaweed, with its rapid growth and high carbon absorption capacity, emerges as a significant contributor to these ecosystems. This study highlights the collaboration between Asosiasi Rumput Laut Indonesia (ARLI) and The China Algae Industry Association (CAIA), showcasing their joint efforts in promoting seaweed as a viable carbon sink. The partnership has achieved notable advancements in understanding the carbon sequestration potential of seaweed and integrating it into carbon trading frameworks.

Lessons learned from this cooperation illustrate both the challenges and successes of implementing seaweed-based carbon projects. By analyzing the specific achievements and lessons learned from this partnership, this study aims to highlight the practical applications of seaweed-based carbon sequestration projects and their potential to drive sustainable development. Ultimately, this paper underscores the importance of leveraging international partnerships and innovative practices in enhancing blue carbon strategies to combat climate change effectively.

**Key words:** Blue Carbon, Seaweed, Carbon Sequestration, Climate Change Mitigation, Carbon Trading, ARLI-CAIA Partnership.